

VELMÈRE ADVANCED AUDIT REPORT

Synthetix Network Proxy (0xc011a73ee8576fb46f5e1c5751ca3b9fe0af2a6f)

VELMERE SECURITY AUDIT REPORT: SYNTHETIX NETWORK PROXY

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: rep_snx_advanced_054 | Tier: ADVANCED | Application Surface: CANONICAL

Created At: 2026-09-10T12:36:16.284Z

VERDICT SUMMARY:

MODERATE RISK (Risk: 34/100 | Quality: 96/100)

Release Decision: PASS | Verification Status: AUTOMATED_ONLY

Confidence Score: 95/100 Evidence Coverage: 96%

EVM State Provenance: Block #18072000 | Bytecode SHA-256: sha256:51c9d81d2... | DETERMINISTIC_REPRODUCIBLE

VELMERE CRYPTOGRAPHIC AUDIT SEAL - SHA-256 MERKLE ROOT

Root: sha256:2d1e101cec6003be...f32ba838936a

[VERIFIED - IMMUTABLE]

Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: /api/audit/report-pdf

Evidence Coverage Matrix: Bytecode 96% | CFG 93% | Functions 97% | Detectors 100% | Formal Verification 86%
Synthetix relies on a multi-contract delegatcall proxy architecture governed by the Spartan Council and collateralized debt pools.

AUDIT OVERVIEW & CONTRACT CONTEXT

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Synthetix Network Proxy
Contract Address: 0xc011a73ee8576fb46f5e1c5751ca3b9fe0af2a6f
Network / Chain ID: Ethereum Mainnet (ID: 1)
Token / Symbol: SNX
Proxy Pattern: ProxyERC20 (Delegatcall Architecture)
Compiler Version: solc 0.4.25

SOURCE CODE VERIFICATION & BYTECODE DECOMPILEATION

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

BASELINE SECURITY FINDINGS

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

PRIVILEGED ROLES, ADMIN KEYS & GOVERNANCE

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Verification of timelock delay parameters and governance controls

* Spartan Council Governance Elected council with Protocol DAO multi-sig VERIFIED

Detailed role map traces all execution paths back to verifiable on-chain controllers.

LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

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Token distribution curve, whale clusters, and verifiable DEX lock proofs

* SNX Collateral Staked	Over \$380,000,000 securing sUSD debt pool	VERIFIED
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Liquidity analysis examines AMM pairing architecture, lock mechanisms, and transaction fee behavior.

ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

* Oracle Dependency	Pricing feeds inspected in bytecode	VERIFIED
* Flash Loan Attack Vector	State manipulation resilience assessed	VERIFIED
* Reentrancy Protection	CEI pattern / Reentrancy guard verified	VERIFIED

STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Deep storage layout slot collision detection across implementation upgrades

* Delegatecall Target Map	18 interconnected implementation contracts verified	VERIFIED
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Symbolic execution of the storage layout demonstrates memory allocation consistency across upgrades.

FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Mathematical Z3 solver invariant verification and bytecode regression analysis

Deployment Status: Verified Active

Formal Verification (Z3 SMT): VERIFIED: 86% proof coverage (QF_LIA, TIMEOUT!=PASS)

Invariant Fuzzing: 100,000 runs executed (0 violations)

Assertion Breakages: 0 detected in verified states

HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

REVIEW STATUS: AUTOMATED ANALYSIS ONLY (AUTOMATED_ONLY)

Attested reviewer evidence and signed non-automated verification findings

Review status: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED. In accordance with Velmore policy, human review claims are NEVER emitted unless verified reviewer evidence exists.

Reviewer State: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED

Analyst: NONE (Independent manual review not commissioned)

Attestation Signature: NONE (No attestation emitted - review not performed)

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